



University of Piraeus

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Security of Wireless and Mobile Networks

Case 2

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Abstract

Transition from monitoring to exploitation through a targeted deauthentication attack in a controlled environment. Initial detection of the target router's physical address (BSSID) and broadcast channel using the airodump-ng tool. Then, injecting fake management messages to spoof the Access Point's identity and forcibly disrupt the targeted device's connection. Final confirmation of the weaknesses of the 802.11 standard and the lack of authentication mechanisms in management messages, highlighting the network's vulnerability to denial-of-service (DoS) attacks.

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Introduction

This paper is a continuation of Use Case 1, where through passive monitoring and analysis of wireless frames (Management, Control and Data frames) the inherent weakness of the 802.11 standard was highlighted, i.e. the absence of authentication mechanisms in management messages. Based on the above finding, Use Case 2 marks the transition from simple motion recording to active intervention (packet injection) in wireless media. The object of the current scenario is the practical exploitation of the aforementioned vulnerability, through the implementation of a targeted disconnection attack in a tightly controlled environment. The process begins with the necessary stage of identification, utilizing the airodump-ng tool to map the network and collect the critical technical parameters (BSSID and broadcast channel), thus preparing the ground for the upcoming injection of fake messages.

Privacy Note: For security and privacy reasons, some sensitive information has been hidden in this paper. In particular, parts of the MAC addresses (BSSIDs and Station IDs) have been hidden in both the text (using "XX:XX:XX:XX") and the screenshots (using black boxes). Also, some of the names (SSIDs) that betray the location or surnames and details of neighboring networks that were randomly recorded have been covered during scanning.

Practical piece of work

At this stage of the exercise, the transition from sniffing to intervention (packet injection) to wireless media takes place. In particular, the process involves sending fake disconnection messages (Deauthentication frames), with the aim of violently interrupting the communication between the router (Access Point) and a connected device (Client). Before the attack is carried out, it becomes necessary to gather two critical information at the identification stage: the physical address (BSSID) of the router and the channel to which it transmits. To extract this data, the airodump-ng tool is used. Through the execution of the command **sudo airodump-ng wlan0**, the network interface is put into a continuous listening mode, allowing the local wireless spectrum to be scanned, neighboring networks to be recorded, and the required technical data of the target network to be accurately identified.

```
(kali@kali)~$ sudo airodump-ng wlan0
[sudo] password for kali:

Ch 13 ][ Elapsed: 18 s ][ 2026-04-04 14:44

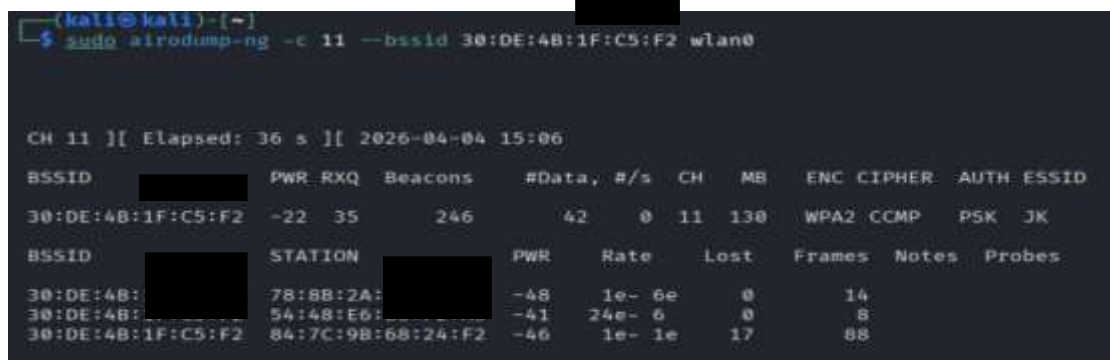
BSSID:          PWR: Beacons:  #Data: #/s  CH:  MB:  ENC CIPHER AUTH ESSID
14:00:0A:D2:19:E53 -70      2      0      0  11  360  WPA2 CCMP PSK COSMOTEL-shdb25
00:1D:1C:FD:4E:29 -76     11      1      0  1  405  WPA2 CCMP PSK Dermentzopoulos
E8:6E:44:15:F9:64 -73      2      0      0  3  130  WPA2 CCMP PSK VODAFONE_H268Q-3372
E0:19:54:4D:03:17 -1       0      0      0  3  -1    WPA2 CCMP PSK <length: 0>
0A:02:10:51:4F:A1:00 -72     16      0      0  4  400  WPA2 CCMP PSK Huawei_XFV4NM
6A:90:14:C1:0E:20:03 -70      3      0      0  13  270  WPA2 CCMP PSK COSMOTEL-734612
7B:8A:20:01:E3:0C -73      1      0      0  6  195  OPN Orestiada WiFi
-73     13      0      0  1  540  WPA2 CCMP PSK COSMOTEL-654980
-70     13      1      0  1  360  WPA2 CCMP PSK 4410
-62     31      0      0  1  405  WPA2 CCMP PSK COSMOTEL_335285_2,4G
-75      0      0      0  1  65   WPA2 CCMP PSK DIRECT-18-HP DeskJet 3830 series
-20     70      6      0  11  130  WPA2 CCMP PSK JK
-32     62      0      0  11  130  WPA2 CCMP PSK <length: 0>
-51     56      0      0  8  130  WPA2 CCMP PSK <length: 0>
-69     14      0      0  11  130  WPA2 CCMP PSK <length: 0>
-68     18      1      0  11  130  WPA2 CCMP PSK Lakis

STATION:          Seen:  Data:  Rate:  Last:  Frames:  Notes:  Probes
30:DE:4B:XX:XX:XX -72      0 - 1e  0      0      1
76:24:22:23:CS:51 -72      0 - 1e  0      0      6
CE:F9:BA:3A:44:38 -74      0 - 1e  0      0     11
70:54:BA:FF:84:38 -62      0 - 1e  0      0      3
AA:46:BD:0C:D8:8C -78      0 - 1e  0      0     12
4E:3E:BA:44:74:68 -72      0 - 1e  0      0     17
D8:1F:12:A5:D1:D7 -70      0 - 1e  0      0     15
-60      0 - 1e  0      0      1
-64      0 - 1e  0      0      1
-58      0 - 1e  0      0      1
-58      0 - 6e  0      0      2
-74      0 - 6e  7      4
-76      0 - 1e  0      0      2
-50     1e- 1e  0      0     14

[not associated]
[not associated]
[not associated]
moduletest
To A34 tou/tric Bayy
```

These results led to the successful identification of the target network, **hel(wifi)** (ID "JK") and the extraction of the critical parameters for the implementation of the attack. In particular, the physical address (BSSID) of the router was recorded as **30:DE:4B:XX:XX:XX**, as well as its operating channel (Channel 11). At the same time, in the logging of connected customers (STATION) of airodump-ng, the MAC address of a device was identified (**84:7C:9B:XX:XX:XX**), which was actively communicating with that router. The correctness of this finding was

also fully verified through the administrator application (TP-Link app) of the router, where the active connection of the specific device to the network was visually confirmed, bearing the same exact MAC address (as shown in the relevant screenshot). In order to optimize the targeting of the attack and avoid packet loss, it was deemed necessary to focus the wireless interface exclusively on the specific channel and network. This configuration was achieved through the execution of the command **sudo airodump-ng -c 11 --bssid 30:DE:4B:XX:XX:XX wlan0**. This process was kept active, now acting as a mechanism for continuously monitoring targeted traffic in real time, in order to visually confirm the successful disconnection of the target device during the upcoming Deauthentication attack.



In order to execute the attack and send the fake Deauthentication frames, it was necessary to start a new session on a second, independent terminal. In this new environment, the aireplay-ng tool was used, which specializes in the process of active injection (packet injection), undertaking the creation and direct transmission of the manufactured packages to the wireless medium. The implementation of the targeted attack was carried out with the execution of the order **sudo aireplay-ng --deauth 15 -a 30:DE:4B:XX:XX:XX -c 84:7C:9B:XX:XX:XX wlan0**. Through this command,

the tool proceeded to send a specified number of fake disconnect packets, pretending to be the identity of the Access Point (based on the -a parameter), with the sole purpose of forcibly interrupting the connection of the default client device (based on the -c parameter).

```
(kali@kali)-[~]
$ sudo aireplay-ng --deauth 15 -a 30:DE:4B:  -c 84:7C:9B:  wlan0
15:09:36 Waiting for beacon frame (BSSID: 30:DE:4B:  ) on channel 11
15:09:37 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [15:33 ACKs]
15:09:38 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [12:34 ACKs]
15:09:38 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 0:33 ACKs]
15:09:39 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 0:51 ACKs]
15:09:40 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 1:40 ACKs]
15:09:41 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 0:36 ACKs]
15:09:42 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [49:69 ACKs]
15:09:43 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 0:50 ACKs]
15:09:44 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 1:59 ACKs]
15:09:45 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 0:64 ACKs]
15:09:46 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [87:88 ACKs]
15:09:47 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [85:85 ACKs]
15:09:48 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 6:58 ACKs]
15:09:48 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [ 0:41 ACKs]
15:09:49 Sending 64 directed DeAuth (code 7). STMAC: [84:7C:9B:  ] [1:41 ACKs]
(kali@kali)-[~]
$
```

Analysis of the terminal log captures the core of the Deauthentication attack in full swing. In particular, the display of the message "**Sending 64 directed DeAuth (code 7)**" indicates that the aireplay-ng tool generates and injects continuous dummy Management frames, which are targeted at the selected device. At the same time, the indication of the confirmations received "[x | y ACKs]" is the most critical finding of the process, as it certifies the interaction of the target with the manufactured packages. The recording of these ACKs proves in practice that the targeted device receives, accepts and processes the forged frames as completely valid and legal, which results in the immediate and violent termination of its connection from the network.

The success of the Deauthentication attack is fully confirmed by the analysis of the tracking data through the airodump-ng tool. In particular, the recording of an extremely high number of lost frames for the target device is a direct symptom of the targeted interference and the consequent collapse of the connection. The most notable finding, however, lies in forcing the device to automatically reconnect after the attack ends. During this process, passive listening succeeded in intercepting the authentication process (4-way handshake), a fact that is visually certified both by the indication "WPA handshake" in the general context, and by the presence of "EAPOL" type packages in the individual customer data. This development highlights in practice how a local denial of service attack can be the critical preparatory stage for the complete breach of a WPA2 network, allowing the collection of the necessary cryptographic data for the further decoding (offline password cracking) of the password.


```
(kali@kali)-[~]
$ sudo airodump-ng -c 11 --bssid 30:DE:4B: wlan0

CH 11 ][ Elapsed: 4 mins ][ 2026-04-04 15:10 ][ WPA handshake: 30:DE:4B:
BSSID PWR RXQ Beacons #Data, #/s CH MB ENC CIPHER AUTH ESSID
30:DE:4B: -23 0 1603 411 0 11 130 WPA2 CCMP PSK JK
BSSID STATION PWR Rate Lost Frames Notes Probes
30:DE:4B: 78:8B:2A: -48 1e- 1e 0 173
30:DE:4B: 54:48:E6: -44 6e- 6e 2 64
30:DE:4B: 84:7C:9B: -46 1e- 1e 337 2398 EAPOL JK
```

Conclusion

In the second and final stage of the exercise, the transition from passive monitoring to active intervention took place, through the successful implementation of a targeted Deauthentication-Packet Injection. By exploiting the critical security hole of the original IEEE 802.11 standard—i.e., the absence of mandatory authentication in management messages—it was possible to spoof the identity of the Access Point and force the disconnection of a legitimate device from the network.

Monitoring the channel during the attack through the airodump-ng tool confirmed the complete collapse of communication, which was reflected in the high number of lost frames. However, the most critical finding of the process was the behavior of the device after the interruption: in an attempt to automatically reconnect, it initiated the 4-Way Handshake process. The monitoring tool successfully captured the authentication packets (EAPOL), intercepting the network's cryptographic access data.

This paper proves in practice that the Deauthentication attack is not just a DoS method to cause malfunctions. Rather, it is the fundamental, strategic first step to a complete wireless network breach (WPA/WPA2), as it forces the target to reveal its handshake to the attacker. These findings underscore the imperative to transition to newer standards and technologies, such as IEEE 802.11w (Protected Management Frames) and WPA3, which effectively shield network "administrative" messages against such spoofing.

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